

ALEX MADARAS

JUNIOR SOFTWARE DEVELOPER

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ABOUT ME

I'm a Computer Science Master's graduate and aspiring Software Engineer with a passion for building meaningful software – especially mobile and web applications that enhance people's travel experiences. Curious and self-driven, I enjoy exploring new technologies and applying them to both personal and professional projects.

Web Portfolio: [alexmadaras00.github.io](#)

EDUCATION

Master's in Computer Science

University of Twente

Enschede, The Netherlands

Graduated in March 2025

Courses: Software Engineering Techniques, Software Management, Requirement Elicitation, Software Architecture, Microservices Architecture in Web Development, Machine Learning, Exploratory Data Analysis.

Bachelor's in Computer Science

West University of Timisoara

Timisoara, Romania

2019–2022

Courses: Algorithms and Data Structures, Introduction to Python, Software Engineering, Practice Stage, Statistics and Numerical Analysis, Web Technologies, Artificial Intelligence, Object-Oriented Programming

Erasmus+ Semester – Bachelor of Game and Apps Development

IADE Creative University

Lisbon, Portugal

2020–2021

Courses: Collaborative Learning, Mobile Programming in Kotlin, Game Frameworks, Agile Management for Multiplatform Projects.

Focus: building a Multiplatform Project consisting of a Windows 2D Game in Unity connected to a Companion App in Android through an SQL Database.

WORK EXPERIENCE

ING Nederland B.V.

AI Engineer Intern

Amsterdam, Netherlands

Apr 2024 – Nov 2024

- **Researched AI algorithms** to detect data quality issues in credit risk data for my Master's thesis (graded 8.0/10).
- Performed **Exploratory Data Analysis** on ING's **credit risk data set**, while improving my business understanding, communication and delivery skills through presentations, including to CIO-level stakeholders.
- Built, trained, and validated an ensemble of one machine learning model (**Isolation Forest**) and a deep learning model (**Autoencoders**) using Keras, identifying 6 suspicious rows, resulting in a Proof of Concept at ING, now continued by another team.

Victory Square Partners

Android Developer Intern

Timisoara, Romania

Jul 2021 – Sept 2021

- Developed practical **Android** skills using **MVVM** architecture within an **Agile Scrum** team, applying **GIT** for effective version control and collaboration.
- Contributed to the migration of a mobile door lock app to an MVVM architecture, while enhancing the UI of **50%** of key screens in the Android app, improving usability and user satisfaction.
- Implemented **Firebase Analytics** to track user behavior, enabling data-driven improvements.

REMARKABLE PROJECTS

Java Windows PC Game – AzulApp

Java Engineer

Enschede, Netherlands

Apr 2023 – June 2023

- Developed a **Java** desktop game in a team of 4 using **TDD**, **Clean Code** principles, and **MVC architecture**.
- Implemented core gameplay logic, a **GitLab CI/CD** pipeline (YAML), and applied 4 design patterns (Singleton, Observer, Proxy, Facade).
- Achieved **96.89%** test coverage with JUnit and earned a **4.4/5** code quality score on **Sigrid**, indicating a high-quality codebase.

Full Stack Travelling WebApp using Microservice Architecture

Team Lead

Enschede, Netherlands

Feb 2024 – Apr 2024

Re-started in Apr 2025

- Built a full-stack web app using **SOA microservices** with **Java** (back-end), **HTML/Thymeleaf/JavaScript** (front-end), and **PostgreSQL**, focusing on modularity and maintainability. The project's back-end was completed during February and April 2024 as part of a course.
- Containerised services with **Docker** for efficient, isolated deployment.
- Integrated external APIs, including **OpenAI**, via **GET/POST** requests for seamless third-party interaction.
- Since April 2025, it continues as a start-up idea as an MVP, where its front-end and back-end are currently under development, to match its' improved architecture.

SKILLS

Programming: Java, Kotlin, Python, SQL, Node.js

Technical: Android, Firebase, GIT, Machine Learning, Data Analysis, Algorithms and Data Structures

Languages: English (C1 Cambridge Certified), Dutch (A1 level), Romanian (Mother tongue)

Soft Skills: Communicative, Problem Solving, Good Listener, Ambitious, Attention to Details, Critical Thinking.

Interests: Software/Mobile Development, AI, Tour Guiding and Tourism, History, Calisthenics, Psychology

EXTRA-CURRICULARS

Hackathons/Workshops: Won 2nd place in the IADE Creative Hackathon 2021, where we had to design and build in **one week** a Mobile application that helps Erasmus students integrate in Lisbon as Erasmus students.

Local Tour Guide: certified since 2021, participated in several activities and volunteered for the 2023 Cultural Capital Events in Timisoara, guiding groups of 30+ tourists from many countries around the world.

Sports: Captain of the football 7v7 team and the volleyball mix team from my University.

Volunteering: Erasmus+ Projects regarding Sustainability (YouRoTrip Vienna 2022), Alternative Learning (GeoFun Latvia 2022), and Empowering Trainers in Non-Formal Education (Varna 2025)